

## Drug Therapy in Pregnancy Specific Considerations

### Introduction

- Medication use in pregnancy and lactation is controversial due to medicolegal and ethical implications.
- Often underemphasized in health professionals' education.
- Clinicians are responsible for ensuring appropriate therapy:
  - Before conception
  - During pregnancy and delivery
  - After delivery
- Treatment for illnesses during pregnancy may differ from those for nonpregnant women.
- Active patient participation is essential in therapy decisions.
- Medication dosing for pregnant women:
  - Often the same as the general population.
  - Some cases require different dosing or medication selection.
- Principles for lactation medication use are similar but not identical to pregnancy.
- Inclusive language used per ACOG and ILCA guidelines:
  - Gendered terms refer to biological sex, not gender.

### Physiology of Pregnancy

- **Pregnancy Survival Rates:**
  - Only ~40–50% of embryos survive.
  - Most losses occur in the first 2 weeks post-fertilization.
  - <20% of miscarriages occur after 12 weeks gestation.
- **Fertilization Process:**
  - Sperm attaches to the egg's *zona pellucida*, rendering it nonresponsive to other sperm.
  - Enzymes allow sperm to penetrate and contact the egg's cell membrane.
  - Sperm and egg membranes combine, forming a zygote.
  - Chromosomes from sperm and egg fuse, creating a single nucleus for cell division.
- **Zygote Journey:**
  - Fertilization occurs in the fallopian tube.
  - Travels to the uterus over 3–4 days while dividing.
  - By day 3: Reaches the uterus.
  - By day 6: Becomes a blastocyst, shedding the *zona pellucida* and implanting in the endometrium.
- **Blastocyst and Early Pregnancy:**
  - Implantation starts ~6 days post-fertilization.
  - hCG detectable in blood or urine 8–9 days after ovulation.
  - By day 10: Fully implanted and nourished by maternal blood.
- **Embryonic to Fetal Transition:**
  - Embryonic period: Weeks 2–8 post-fertilization.
    - Most body structures form during this phase.

- Fetal period: Begins after week 8 and lasts until term (~40 weeks from the last menstrual period).
  - Growth and maturation continue during this phase.
- **Gravidity and Parity:**
  - Gravidity: Number of pregnancies, with multiple births counting as one.
  - Parity: Number of pregnancies exceeding 20 weeks of gestation.
  - Medical abbreviations for G (gravidity) and P (parity):
  - Reflect term deliveries, premature deliveries, aborted pregnancies, and living children.
  - Example: G4P2-1-1-3 (4 pregnancies, 2 term deliveries, 1 premature delivery, 1 ectopic pregnancy, 3 living children).

### **Characteristics of Pregnancy:**

- Duration: Approximately 280 days (40 weeks or 9 months)
- Measured from the first day of the last menstrual period (LMP) to birth
- Gestational age: Starts from the first day of the LMP (2 weeks prior to fertilization)
- Estimated due date: Add 7 days to the first day of the LMP, then subtract 3 months
- Divided into three trimesters:
  - 1st Trimester: Weeks 1-13
  - 2nd Trimester: Weeks 14-27
  - 3rd Trimester: Weeks 28-40 (delivery)
- **Symptoms of Pregnancy:**
  - Fatigue
  - Increased frequency of urination
  - Nausea and vomiting (morning sickness), typically resolves by 14-16 weeks
  - Fetal movement (first felt at 13-16 weeks, most by 18-20 weeks)
- **Signs of Pregnancy:**
  - Cessation of menses
  - Change in cervical mucus consistency
  - Bluish discoloration of the vaginal mucosa
  - Increased skin pigmentation
  - Anatomical breast changes

### **Pharmacokinetic Changes During Pregnancy**

- Physiologic changes begin in the first trimester and peak during the second trimester.
- Increased plasma volume, cardiac output, and glomerular filtration (50%, 30%-50%, and 50%-80%, respectively).
- These changes may lower the concentration of renally eliminated medications.
- Changes in plasma volume and body weight can affect the volume of distribution of medications.
- Plasma albumin concentration decreases, increasing the volume of distribution of protein-bound medications.

- Unbound medications are cleared more rapidly by the liver and kidney, resulting in little change in concentration.
- Hepatic perfusion increases, potentially increasing hepatic extraction of medications.
- Activity of metabolic enzymes and drug transporters changes:
  - Increased activity of cytochrome P450 3A4, 2C9, 2D6, and UGT1A4.
  - Decreased activity of cytochrome P450 1A2.
- Nausea, vomiting, and delayed gastric emptying may alter medication absorption.

## **Transplacental Medication Transfer**

- The placenta is the organ of exchange between the pregnant individual and fetus, including medications.
- Medications cross the placenta mainly through diffusion.
- Chemical properties influencing transplacental transfer:
  - Half-life, lipid solubility, electrical charge, molecular weight, and protein binding.
- Preference for medications with:
  - High molecular weight, short half-life, and high protein binding (lower potential to cross the placenta).
- Molecular weight and placental transfer:
  - Molecules <500 Da cross easily.
  - Molecules 600-1,000 Da cross slowly.
  - Molecules >1,000 Da (e.g., insulin, heparin) do not cross significantly.
- Monoclonal antibodies, despite their high molecular weight, cross the placenta and may have higher infant levels than maternal levels at delivery.
- Lipophilic medications (e.g., opioids, antibiotics) cross the placenta more easily than water-soluble medications.
- Decreased plasma albumin may lead to higher concentrations of protein-bound medications in the fetus.
- Fetal pH is slightly more acidic than maternal pH:
  - Weak bases (e.g., bupivacaine) become ionized, leading to ion trapping and higher fetal levels, less likely to diffuse back into maternal circulation.

## **PHARMACOTHERAPY SELECTION DURING PREGNANCY**

- **Baseline risk for congenital malformations:**
  - Approximately 3% to 5%.
  - Medication exposure accounts for less than 1% of birth defects.
  - 80% of birth defects are of unknown causes, 15% are chromosomal, 4% genetic, and less than 1% are due to other environmental factors.
- **Factors influencing medication outcomes:**
  - Stage of pregnancy during exposure.
  - Medication route, administration, and dosage.
- **Impact of teratogen exposure:**

- **First 4 weeks:** Risk of “all-or-none” outcomes (either embryo destruction or no effect).
- **Embryonic period (organogenesis):** Teratogens may cause structural anomalies.
- **Later pregnancy:** Risk of growth restriction, CNS abnormalities, impaired organ function, fetal demise.
- **Examples of teratogenic medications:**
  - **Embryonic period:** Chemotherapy agents (e.g., methotrexate, cyclophosphamide), sex hormones (e.g., androgens), lithium, retinoids, thalidomide, warfarin, certain antiseizure meds.
  - **Second/third trimester:** NSAIDs, tetracycline derivatives.
- **Medication use during pregnancy:**
  - Necessary for treating acute and chronic conditions.
  - Minimize exposure to harmful medications by eliminating nonessential drugs, discouraging self-medication, and adjusting doses.

## **Preconception Planning**

- Pregnancy outcomes are affected by the individual's health, lifestyle, and history before conception.
- Goal: Health promotion through modification of behavioral, biomedical, and social risks in individuals of reproductive age who can become pregnant.
- Aim: Ensure optimal health and improve pregnancy outcomes.
- Preconception care is crucial as certain behaviors and exposures pose risks to the fetus during the first trimester, often before prenatal care begins or pregnancy is detected.
- **Common Major Congenital Abnormalities:**
  - Neural tube defects (NTDs), cleft palate, lip, and cardiac anomalies.
  - Approximately 3,000 pregnancies in the U.S. are affected by NTDs annually.
- **Folic Acid Supplementation:**
  - Significantly reduces NTD incidence.
  - Recommended: 0.4 mg daily during reproductive years for individuals of childbearing potential.
  - Higher dose (4 mg daily) recommended for those at high risk (personal or family history of NTDs).
  - Start supplementation 3 months before conception and continue until 12 weeks gestation.
- **Use of Alcohol and Substances During Pregnancy:**
  - Risk factors include alcohol, substance, and cigarette use during pregnancy.
  - Risks associated with these substances should be addressed in preconception care.
- **Nicotine Replacement Therapy:**

- Recommended for pregnant individuals trying to quit smoking, though potential risks of nicotine use should be considered.
- Less risky than cigarettes containing over 4,000 chemicals.
- Bupropion has no pregnancy risk but has limited efficacy for smoking cessation.
- Varenicline has limited safety data in pregnancy but no teratogenic effects shown.

**TABLE 99-1 Selected Preconception Risk Factors for Adverse Pregnancy Outcomes**

| Preconception Risk Factor                                                   | Potential Adverse Pregnancy Outcomes                                                                                                                                                                                                                     | Management or Prevention Options                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use of known teratogens</b>                                              |                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>• Antiseizure Medications</li> </ul> | <ul style="list-style-type: none"> <li>• Some are known teratogens; increased risk for craniofacial, NTD, cardiac, and limb defects<sup>a</sup></li> <li>• Fetal hydantoin syndrome</li> </ul>                                                           | <ul style="list-style-type: none"> <li>• Optimize to lower risk therapy while maintaining control prior to conception</li> <li>• Avoid valproic acid</li> <li>• Use monotherapy if possible</li> <li>• Start folic acid at least 0.4 mg daily preferably at least 1 months prior to conception</li> </ul>                                                                                           |
| <ul style="list-style-type: none"> <li>• Retinoids</li> </ul>               | <ul style="list-style-type: none"> <li>• Spontaneous abortion</li> <li>• Known teratogen; increased risk for CNS, craniofacial, and cardiac defects<sup>a</sup></li> </ul>                                                                               | <ul style="list-style-type: none"> <li>• Discontinue at least 1 month (isotretinoin, bexarotene) or 3 years (acitretin) before attempting conception</li> <li>• Enrolled in iPLEDGE (isotretinoin), Education and Pregnancy Prevention for Acitretin (EPPA) or Do Your P.A.R.T. (acitretin). Follow manufacturer recommendations for contraception (isotretinoin, acitretin, bexarotene)</li> </ul> |
| <ul style="list-style-type: none"> <li>• Warfarin</li> </ul>                | <ul style="list-style-type: none"> <li>• Fetal warfarin syndrome</li> </ul>                                                                                                                                                                              | <ul style="list-style-type: none"> <li>• Switch to nonteratogenic anticoagulant (eg, LMWH) before becoming pregnant</li> <li>• Patients with mechanical valves may remain on warfarin for some or all of the pregnancy</li> </ul>                                                                                                                                                                   |
| <b>Lifestyle factors</b>                                                    |                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>• Alcohol use</li> </ul>             | <ul style="list-style-type: none"> <li>• Fetal alcohol syndrome</li> </ul>                                                                                                                                                                               | <ul style="list-style-type: none"> <li>• Cease alcohol intake before conception</li> </ul>                                                                                                                                                                                                                                                                                                          |
| <ul style="list-style-type: none"> <li>• Obesity</li> </ul>                 | <ul style="list-style-type: none"> <li>• Malformations (eg, NTD and orofacial)</li> <li>• Preterm delivery</li> <li>• Spontaneous abortion</li> <li>• Stillbirth</li> <li>• Macrosomia</li> <li>• Impaired growth</li> <li>• Cesarean section</li> </ul> | <ul style="list-style-type: none"> <li>• Weight loss with appropriate nutritional intake before pregnancy</li> </ul>                                                                                                                                                                                                                                                                                |

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|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Tobacco use</li> </ul>  | <ul style="list-style-type: none"> <li>• Preterm birth</li> <li>• Low birth weight</li> <li>• Spontaneous abortion</li> <li>• Increased perinatal mortality</li> <li>• Orofacial clefts</li> <li>• Intrauterine growth restriction</li> <li>• Sudden infant death syndrome</li> </ul> | <ul style="list-style-type: none"> <li>• Ideally, cease tobacco use before conception</li> <li>• Individualize approach with nonpharmacologic therapies (including psychosocial and behavioral interventions like CBT, motivational interviewing, and counseling) and can consider NRT products, bupropion, varenicline. Discuss risks and benefits with the patient</li> </ul> |
| <ul style="list-style-type: none"> <li>• Cannabis use</li> </ul> | <ul style="list-style-type: none"> <li>• Intrauterine fetal demise</li> <li>• Low birth rate<sup>b</sup></li> <li>• Preterm delivery<sup>b</sup></li> </ul>                                                                                                                           | <ul style="list-style-type: none"> <li>• Ask about medical and nonmedical use of marijuana in all patients</li> <li>• Educate about potential risks</li> <li>• Encourage to discontinue use ideally prior to conception</li> </ul>                                                                                                                                              |

<sup>a</sup>List is not all-inclusive.

<sup>b</sup>When stratified by frequency of use (risk at least weekly use).

CBT, cognitive behavioral therapy; CNS, central nervous system; HTN, hypertension; NRT, nicotine replacement therapy; NTD, neural tube defect; VTE, venous thromboembolism.